

SoC is not just a display, but a complete embedded system on its own, an example specification would help you get the idea. Here is a sample of such a specification: <http://bit.ly/18QF8jf> The page also contains links to other chipsets which have been used in different smartphones. Go through them to understand what goes into the making of a smartphone (but that's another story).

An embedded system can't be too flexible in terms of display resolution and frame rates, for both of which the designers must put limits before actual assembling or production takes place. Taking the example of MediaTEK MT6575 (for which we've provided the link earlier), you can't attach a 1080p screen to the chip; mainly because the processor, system bus and memory aren't built for that.

A display or screen needs signals for every frame to be fed to it. The signals are fed by the display adapter which is attached to the system. The display adapter has two requirements of its own from the system – supply of display data, which is to be converted into signals for the display and power consumption for signal processing. Display data is usually kept in a part of the main system memory (this is analogous to a PC with an on-board GPU) or in a dedicated graphics memory (and this is analogous to having a discreet graphics card in a PC). The bigger the display is, the more power and memory it requires.

When it comes to graphics controllers, you have two options:

- 1 Basic graphics controller: The graphics memory is shared with the main memory of the system. This option is good for lower resolutions with less power hungry requirements.
- 2 Dedicated graphics controller (such as an NVIDIA Tegra chip): The graphics memory is separated as is the computation. This is good for bigger displays which have more power requirements.

Communications and networking

Home appliances are one of the most widespread uses of embedded systems. Many of them don't need any networking features, but all that's about to change as we progress towards a more connected home. While you wait for that to happen, let's take a look at the various applications of embedded systems in devices facilitating communications, and the challenges associated with them.

Mobile phones

Most mobile phones have the communication systems connected on the main board. While integrating them is a challenge of its own, there's no major effort needed these days, as SoC designs that do this are already flourishing.